Benchmarking General and Medical Small Language Models for Spanish Obstetric Question Answering: The Impact of Fine-Tuning

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***Abstract*—Small Language Models (SLMs) offer a computationally efficient alternative to large language models for clinical question answering; however, evidence on domain adaptation for Spanish obstetric applications remains limited. This study compares the general-purpose Gemma 4 model and the medically pretrained MedGemma 1.5 before and after grounded QLoRA adaptation using MaternaQA-es, while examining No RAG, Hybrid retrieval, and independent HyDE as experimental factors. All 24 scenario-condition runs were evaluated across two 10-question scenarios, yielding 240 final condition-item outputs. Within this exploratory evaluation, on MaternaQA-es, Gemma 4 Base with Hybrid retrieval achieved the highest observed Answer Correctness (0.649), while Gemma 4 QLoRA with No RAG and Hybrid retrieval achieved the highest Answer Relevancy (0.776) and Semantic Similarity (0.838), respectively. On the obstetric challenge set, Gemma 4 Base with Hybrid retrieval achieved the highest Answer Correctness (0.457), MedGemma 1.5 QLoRA with No RAG the highest Answer Relevancy (0.913), and Gemma 4 Base and Gemma 4 QLoRA with HyDE tied in Semantic Similarity (0.707). HyDE achieved higher RAGAS Context Precision than Hybrid in seven of eight model-scenario comparisons; effects on answer quality were heterogeneous and model-dependent. Under No RAG, QLoRA increased Answer Relevancy in all four model-scenario comparisons, while changes in Answer Correctness and Semantic Similarity were mixed. Given the limited evaluation of 10 questions per scenario, these findings should be interpreted as preliminary descriptive patterns.**

***Index Terms*—Small language models, QLoRA, retrieval-augmented generation, HyDE, obstetric question answering, Spanish clinical NLP, model benchmarking.**

I. Introduction

The rapid adoption of Large Language Models (LLMs) has transformed the development of clinical decision-support systems and medical question-answering (QA) applications. Nevertheless, their computational requirements limit deployment in many healthcare environments, motivating increasing interest in Small Language Models (SLMs), which offer lower inference costs, reduced hardware requirements, and easier local deployment while maintaining competitive performance in domain-specific tasks [1], [2]. In medical applications, supervised fine-tuning has emerged as a practical strategy for adapting compact language models to specialized clinical domains while preserving computational efficiency.

Recent studies have shown that domain-specific fine-tuning and retrieval can improve medical question answering under resource constraints [3]. Work on fine-tuned clinical embedding models likewise demonstrates the value of domain-specific retrieval, but remains outside Spanish obstetric QA [4]. Recent obstetric applications include simulated-case decision support, emergency-response evaluation, and neonatal growth prediction [5]–[7]. Reviews of maternal-health RAG emphasize the lack of standardized benchmarks and clinically validated evaluation protocols [8]. This gap is especially relevant in obstetric care, where inaccurate recommendations may affect maternal and fetal safety [9].

An important question therefore remains: does medical pretraining continue to provide measurable advantages after domain-specific fine-tuning? Medically pretrained SLMs should encode biomedical terminology, clinical concepts, and reasoning patterns. Yet, after both a general-purpose and a medical model receive identical specialized supervision, it remains unclear whether prior medical knowledge provides an additional benefit or whether the general-purpose model can achieve comparable or superior performance.

To investigate this question, this study uses MaternaQA-es, a Spanish obstetric QA resource containing 5,727 synthetic question-answer pairs generated from 63 Spanish-language clinical documents. The dataset is publicly available through Hugging Face [10]. The benchmark compares Gemma 4 and MedGemma 1.5 before and after identical grounded QLoRA adaptation.

This work makes three contributions. First, it compares general-purpose and medically pretrained SLMs for Spanish obstetric QA. Second, it evaluates the effect of domain-specific QLoRA adaptation under matched training conditions. Third, it provides a factorial benchmark incorporating retrieval-augmented inference and item-level evaluation.

II. Methodology

A. Study Design

The controlled, offline factorial benchmark crosses model family (Google Gemma 4 E2B instruction-tuned or MedGemma 1.5 4B instruction-tuned), adaptation (Base or

TABLE I
PRESPECIFIED FACTORIAL EXPERIMENT MATRIX

ID	Model family	Adaptation	Retrieval
C01	Gemma 4	Base	No RAG
C02	Gemma 4	Base	Hybrid
C03	Gemma 4	Base	HyDE
C04	Gemma 4	QLoRA	No RAG
C05	Gemma 4	QLoRA	Hybrid
C06	Gemma 4	QLoRA	HyDE
C07	MedGemma 1.5	Base	No RAG
C08	MedGemma 1.5	Base	Hybrid
C09	MedGemma 1.5	Base	HyDE
C10	MedGemma 1.5	QLoRA	No RAG
C11	MedGemma 1.5	QLoRA	Hybrid
C12	MedGemma 1.5	QLoRA	HyDE

Note. *Base* denotes published weights without an adapter; *QLoRA* denotes the grounded adapter; *No RAG* denotes no retrieved context.

grounded QLoRA), and retrieval (No RAG, canonical Hybrid, or canonical independent HyDE), yielding 12 conditions (Table I). "Base" means published weights without an adapter; "No RAG" means answering the question without retrieved context. These factors are independent. Retrieval comparisons hold the answer model, adaptation state, and items fixed. Retained artifacts cover all 24 scenario-condition runs across two scenarios.

B. Data, Scenarios, and Split Integrity

MaternaQA-es, publicly available on Hugging Face [10], contains 5,727 Spanish obstetric QA pairs from 63 clinical documents. Document-level partitions comprise 52 training PDFs (5,093 items), 2 validation PDFs (306 items), and 3 held-out test PDFs (328 items), with no test-source overlap with training or validation. Evaluation uses the first 10 held-out test records, a nonrandom subset rather than the complete partition.

The second scenario uses 10 obstetric challenge questions (`datasets/sample10.jsonl`). Each scenario shares the same 10 `qa_id` values across all conditions. Both support reference-answer metrics. MaternaQA-es retains QA metadata and source-reference information. The challenge items record only QA-level provenance (workbook, sheet, and row); no corpus chunk identifier is stored, although their reference answers match corpus chunks by text.

The 2,268-chunk retrieval corpus includes source material associated with the held-out questions. Thus, RAG tests evidence use when relevant sources are available, not strict generalization to unseen documents. Inference-time corpus access is distinct from parameter leakage: test documents were not used in fine-tuning. Scenarios are complementary, not directly comparable samples: input construction and source-context availability confound absolute between-scenario differences. All principal comparisons are within scenario. In particular, absolute metric differences between MaternaQA-es and the challenge set are not interpreted as effects of fine-tuning or retrieval. The benchmark instead compares model and retrieval configurations on the same questions within each evaluation setting.

C. Models and Grounded QLoRA

QLoRA conditions load `iue-edu/MaternaCare-ES-gemma4-qlora` or `iue-edu/MaternaCare-ES-medgemma-qlora` on the corresponding base model. Base identifiers and shared 4-bit QLoRA settings [11] appear in Table II. Both families use identical training and validation partitions, optimization settings, and `sft_grounded` inputs. This conversational `Contextofuente` representation concatenates pre-associated source context with each training question, unlike inference-time retrieval. Training context is supplied with each example, whereas RAG selects evidence from the separate corpus for each question. This distinction separates grounded parameter adaptation from the retrieval factor evaluated at inference. Fine-tuning and local neural-model computations used an NVIDIA GeForce RTX 5070 Ti Laptop GPU with 12 GB dedicated VRAM.

Fig. 1 contrasts full fine-tuning with LoRA's frozen-base, trainable low-rank branch. The benchmark uses QLoRA with a frozen quantized base; full fine-tuning is not an experimental arm.

D. Retrieval

RAG [12] uses curated obstetric documents. Fig. 2 summarizes the retrieval alternatives, not the complete prompt or runtime graph. The corpus-filtering `clinical_score` adds one point per clinical term (two for multiword terms) and one per action term, testing substring presence in normalized text rather than occurrence frequency. Cumulative bonuses add two points at each Unicode word-count threshold of 120

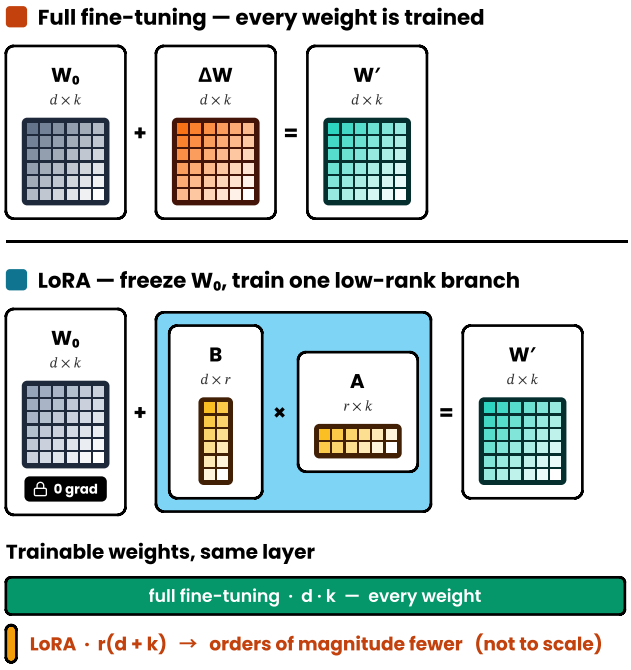


Fig. 1. Conceptual full fine-tuning versus LoRA. QLoRA uses the frozen-base approach with quantized base weights; full fine-tuning was not evaluated in this benchmark.

TABLE II
GROUNDED QLORA CONFIGURATION USED FOR BOTH MODEL FAMILIES

Parameter	Shared setting
Base model	Gemma 4 E2B: `google/gemma-4-E2B-it` MedGemma 1.5 4B: `google/medgemma-1.5-4b-it`
Training representation	sft_grounded
Train / validation examples	5,093 / 306
Quantization	4-bit NF4, double quantization
LoRA rank (r)	16
LoRA alpha	16
LoRA dropout	0.05
Target modules	All linear (all-linear)
Learning rate	2×10^{-4}
Batch size	1 per device
Gradient accumulation steps	8
Effective batch size	8
Epochs / training steps	2 / 1,274
Optimizer	AdamW 8-bit
Learning-rate scheduler	Cosine
Maximum sequence length	1,024 tokens
Packing	Disabled
Random seed	3407

and 350. A reference-line ratio of at least 0.35 subtracts three points; the final score is floored at zero. The ratio counts reference-matching nonempty lines over all nonempty lines. Normalization applies Unicode normalization, removes combining marks, lowercases text, replaces non-ASCII letters and punctuation with spaces, and collapses whitespace.

Dense retrieval uses `BAAI/bge-m3` [13] with SentenceTransformer-normalized vectors on CPU (`batch_size=32`) in a persistent local LanceDB table [14]. Search is exhaustive exact cosine search, without an approximate index; default depth is five. Sparse retrieval uses Okapi BM25 (`rank_bm25/BM25Okapi`) over Unicode lowercase word tokens. Outputs are recorded before answer generation for separate retrieval-quality and latency analysis.

Hybrid retrieval fuses dense and BM25 rankings by reciprocal rank fusion with `rank_constant=60` [15]. Candidate depth is the greater of four times the requested final depth and 50, capped by corpus size. Stable chunk IDs break ties; the final five chunks are supplied to the answer model.

HyDE [16] independently generates a Spanish clinical passage from each question using OpenAI `gpt-5-mini-2025-08-07`, separate from the answer model. Each passage is generated once, cached across answer-model conditions, and embedded with BGE-M3 for exact cosine search. Only retrieved corpus chunks, never hypothetical text, provide answer evidence [17]. Cache namespaces separate dataset mode, provider, and generator model; reuse checks `qa_id` and a question hash.

E. Inference and Reproducibility

Spanish answer prompts use the question alone before retrieval. RAG adds a delimited context block with positional labels; No RAG retains the same question formatting and chat template without that block. Provenance stays in benchmark

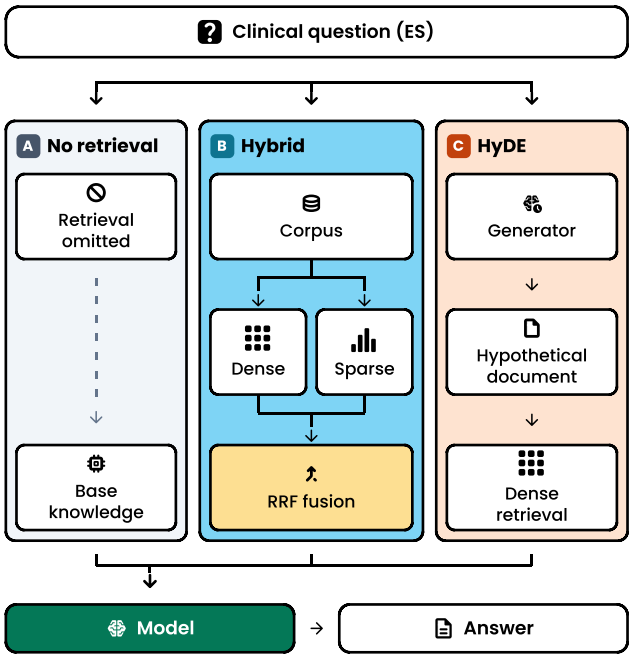


Fig. 2. Retrieval alternatives: No RAG, Hybrid, and independent HyDE.

records, not prompt content. Retrieval and answer generation remain separate. Inference uses the model's chat template, greedy decoding, and at most 2,048 new tokens; sampling is disabled unless a run records otherwise. The same question formatting is retained across retrieval conditions, so No RAG differs by omission of the retrieved-context block rather than by receiving additional source information in the question.

Reproducibility requires dataset revision, corpus hash, complete configuration, prompts, model/adapter identifiers, HyDE provider settings and cache identity, evaluator version, decoding settings, pinned BGE-M3 revision, and deterministic RRF ties.

F. Metrics and Statistical Analysis

RAGAS [18] provides six metrics (Table III): Context Precision, Context Recall, Faithfulness, Answer Relevancy, Answer Correctness, and Semantic Similarity. Context Recall assesses coverage of the reference answer by retrieved contexts; it does not require gold chunk identifiers. Reference-chunk identity checks are separate. Context-dependent metrics are undefined for No RAG, not assigned zero. Response quality, retrieval quality, and operational measurements are analyzed separately.

Evaluator settings are fixed in the evaluation code: RAGAS 0.4.3, with `gpt-5.4-mini` as the LLM judge, `text-embedding-3-small` for embedding-based metrics, and a 2048-token completion limit per evaluator call.

Operational variables include input, output, and total tokens; retrieval, generation, and end-to-end latency; and output tokens per generation second. This wall-clock rate is descriptive, not hardware-independent throughput.

Analysis is exploratory and descriptive, averaging the 10 question-level scores per applicable metric and scenario. Paired contrasts use `qa_id`: QLoRA minus Base within family and retrieval strategy, retrieval contrasts within family and adaptation, and model-family contrasts under matched conditions. For a fixed question, scenario, model family, and metric, the interaction is the QLoRA-minus-Base difference under Hybrid minus that under No RAG. Analogous contrasts compare HyDE with No RAG and the two model families. These are exploratory contrasts, not significance tests.

Approximate marginal 95% Student-t confidence intervals use the exported mean plus or minus the two-sided critical value with nine degrees of freedom times the standard error. For 10 questions, that error is the exported sample standard deviation divided by the square root of 10. This independent-question approximation is limited by nonrandom selection and shared source documents. Confidence intervals for all metrics are derivable from the accompanying sample standard deviations, and unstandardized mean differences accompany the reproducibility files. Paired tests, bootstrap intervals, and standardized paired effects were not performed and are not reported.

III. Results

A. Available Runs and Analysis Set

Table III reports the mean metrics for both scenarios; Table IV reports Answer Correctness confidence intervals.

The final analysis comprised 240 condition-item outputs: 120 for the MaternaQA-es subset and 120 for the obstetric challenge set. Each scenario contained 10 questions evaluated across all 12 conditions, yielding 10 final responses per condition. All automatic metrics applicable to each condition were available; no scores were imputed.

B. MaternaQA-es Subset

Gemma 4 Base with Hybrid retrieval achieved the highest observed mean Answer Correctness (0.649), whereas Gemma 4 QLoRA with No RAG achieved the highest Answer Relevancy (0.776), and Gemma 4 QLoRA with Hybrid retrieval achieved the highest Semantic Similarity (0.838). Under No RAG, QLoRA changed Gemma 4 Answer Correctness, Answer Relevancy, and Semantic Similarity by +0.012, +0.080, and +0.022, respectively. For MedGemma 1.5, the corresponding changes were −0.058, +0.072, and +0.006.

Relative to No RAG, Hybrid and HyDE increased Answer Correctness and Semantic Similarity in all four model-state combinations. Answer Relevancy increased in two of four combinations with Hybrid and one of four with HyDE. Compared directly with Hybrid, HyDE increased Answer Correctness in one of four combinations, Answer Relevancy in two, and Semantic Similarity in two, indicating heterogeneous effects across models and metrics.

C. Obstetric Challenge Set

Within the evaluated obstetric challenge subset, Gemma 4 Base with Hybrid retrieval achieved the highest observed mean Answer Correctness (0.457), and MedGemma 1.5 QLoRA with No RAG achieved the highest Answer Relevancy (0.913). Gemma 4 Base and Gemma 4 QLoRA with HyDE tied for the highest Semantic Similarity (0.707). Under No RAG, QLoRA changed Answer Correctness, Answer Relevancy, and Semantic Similarity by −0.154, +0.129, and −0.017 for Gemma 4, and by −0.054, +0.081, and +0.017 for MedGemma 1.5.

Relative to No RAG, Hybrid increased Answer Correctness in all four model conditions but reduced Answer Relevancy in every condition. Semantic Similarity increased for both Gemma 4 conditions and MedGemma 1.5 QLoRA, but decreased for MedGemma 1.5 Base.

Relative to No RAG, HyDE increased Answer Correctness for Gemma 4 QLoRA (+0.194) and MedGemma 1.5 QLoRA (+0.082), but reduced it for Gemma 4 Base (−0.024) and MedGemma 1.5 Base (−0.028). Answer Relevancy decreased in all four conditions, while Semantic Similarity increased in all four.

Compared directly with Hybrid retrieval, HyDE improved Answer Correctness for Gemma 4 QLoRA (+0.050) and Answer Relevancy for Gemma 4 Base (+0.104) and MedGemma 1.5 Base (+0.177). Semantic Similarity increased in three of four model conditions. These descriptive patterns show that higher retrieval scores did not guarantee uniformly better response metrics.

D. Retrieval-Grounding Metrics

In the MaternaQA-es subset, Hybrid Context Precision ranged from 0.908 to 0.950, Context Recall from 0.800 to 0.900, and Faithfulness from 0.883 to 0.967. HyDE Context Precision was 0.950 in all four model states, Context Recall ranged from 0.750 to 0.900, and Faithfulness from 0.842 to 0.954. HyDE exceeded the corresponding Hybrid Context Precision in three of four model states and increased Context Recall in two of four.

In the obstetric challenge set, Hybrid Context Precision ranged from 0.775 to 0.842, Context Recall from 0.400 to 0.500, and Faithfulness from 0.732 to 0.966. HyDE Context Precision ranged from 0.933 to 1.000, Context Recall was 0.200 in all four model states, and Faithfulness ranged from 0.771 to 0.919. HyDE achieved higher Context Precision than Hybrid for all four model states, while Context Recall was lower for all four.

Across both scenarios, the 20 unique uncached HyDE generations averaged 18.5 seconds, 67.1 input tokens, and 1,437.3 API output tokens. The same hypothetical documents were then reused from cache across the remaining answer-model conditions, reducing model-dependent variation in the query-expansion stage.

E. Uncertainty of Answer Correctness

Table IV reports broad approximate marginal 95% confidence intervals for Answer Correctness, indicating substantial item-level variability. They do not quantify paired differences, and overlap is not a significance test. Rankings remain exploratory; statistically significant superiority is not established.

TABLE III
MEAN AUTOMATIC METRICS FOR THE 10 SHARED QUESTIONS IN EACH SCENARIO

Model	State	Strategy	Answer Correctness	Answer Relevancy	Semantic Similarity	Faithfulness	Context Precision	Context Recall
Panel A. MaternaQA-es test subset								
Gemma 4	Base	No RAG	0.372	0.696	0.704	—	—	—
Gemma 4	Base	Hybrid	**0.649**	0.669	0.820	**0.967**	0.908	0.800
Gemma 4	Base	HyDE	0.560	0.685	0.796	0.954	**0.950**	0.850
Gemma 4	QLoRA	No RAG	0.384	**0.776**	0.726	—	—	—
Gemma 4	QLoRA	Hybrid	0.605	0.759	**0.838**	0.883	0.908	0.800
Gemma 4	QLoRA	HyDE	0.637	0.639	0.818	0.882	**0.950**	0.750
MedGemma 1.5	Base	No RAG	0.388	0.611	0.714	—	—	—
MedGemma 1.5	Base	Hybrid	0.599	0.648	0.794	0.897	**0.950**	**0.900**
MedGemma 1.5	Base	HyDE	0.595	0.737	0.807	0.842	**0.950**	**0.900**
MedGemma 1.5	QLoRA	No RAG	0.330	0.683	0.720	—	—	—
MedGemma 1.5	QLoRA	Hybrid	0.609	0.700	0.826	0.883	0.908	0.850
MedGemma 1.5	QLoRA	HyDE	0.574	0.674	0.828	0.859	**0.950**	**0.900**
Panel B. Obstetric challenge set								
Gemma 4	Base	No RAG	0.359	0.716	0.651	—	—	—
Gemma 4	Base	Hybrid	**0.457**	0.554	0.689	**0.966**	0.800	0.400
Gemma 4	Base	HyDE	0.335	0.658	**0.707**	0.902	0.958	0.200
Gemma 4	QLoRA	No RAG	0.205	0.845	0.634	—	—	—
Gemma 4	QLoRA	Hybrid	0.349	0.736	0.693	0.838	0.775	0.400
Gemma 4	QLoRA	HyDE	0.399	0.672	**0.707**	0.919	0.933	0.200
MedGemma 1.5	Base	No RAG	0.287	0.832	0.631	—	—	—
MedGemma 1.5	Base	Hybrid	0.402	0.368	0.622	0.850	0.842	**0.500**
MedGemma 1.5	Base	HyDE	0.260	0.545	0.639	0.850	**1.000**	0.200
MedGemma 1.5	QLoRA	No RAG	0.234	**0.913**	0.649	—	—	—
MedGemma 1.5	QLoRA	Hybrid	0.455	0.787	0.678	0.732	0.842	0.400
MedGemma 1.5	QLoRA	HyDE	0.316	0.725	0.668	0.771	**1.000**	0.200

Note. Means are calculated from 10 responses per condition and scenario for each applicable metric. Dashes indicate metrics not applicable without retrieval. Bold values identify each metric's maximum within a scenario.

TABLE IV
ANSWER CORRECTNESS: MEAN AND APPROXIMATE MARGINAL 95% CONFIDENCE INTERVAL

Model and state	No RAG	Hybrid	HyDE
MaternaQA-es test subset			
Gemma 4 Base	0.372 [0.238, 0.506]	0.649 [0.456, 0.841]	0.560 [0.375, 0.744]
Gemma 4 QLoRA	0.384 [0.269, 0.500]	0.605 [0.422, 0.788]	0.637 [0.441, 0.833]
MedGemma 1.5 Base	0.388 [0.258, 0.517]	0.599 [0.398, 0.801]	0.595 [0.387, 0.803]
MedGemma 1.5 QLoRA	0.330 [0.213, 0.446]	0.609 [0.454, 0.764]	0.574 [0.401, 0.747]
Obstetric challenge set			
Gemma 4 Base	0.359 [0.159, 0.558]	0.457 [0.244, 0.671]	0.335 [0.125, 0.545]
Gemma 4 QLoRA	0.205 [0.127, 0.282]	0.349 [0.185, 0.513]	0.399 [0.254, 0.544]
MedGemma 1.5 Base	0.287 [0.164, 0.411]	0.402 [0.139, 0.664]	0.260 [0.123, 0.396]
MedGemma 1.5 QLoRA	0.234 [0.152, 0.315]	0.455 [0.249, 0.660]	0.316 [0.098, 0.533]

Note. Mean [approximate marginal 95% confidence interval]; $n = 10$ questions. Student-t intervals assume independent questions; they do not test paired differences.

IV. DISCUSSION

Across the evaluated 20-question benchmark, the observed effect of QLoRA under No RAG varied by model family and scenario. Answer Relevancy increased in all four model-scenario comparisons. Answer Correctness decreased for both model families in the obstetric challenge set and for MedGemma 1.5 in the MaternaQA-es subset; Semantic Similarity decreased for Gemma 4 in the challenge set but increased for MedGemma 1.5 and for both models in MaternaQA-es.

Within the evaluated questions, medical pretraining did not show a uniform advantage after domain-specific adaptation across the tested metrics and configurations. These findings support evaluating multiple dimensions rather than treating a single automatic score as a complete measure of quality.

Hybrid retrieval increased Answer Correctness in all four model conditions in each scenario but reduced Answer Relevancy in six of eight comparisons. One plausible interpretation is that retrieved evidence increased agreement with reference answers while sometimes reducing direct focus. Retrieval-grounding scores were generally high for Context Precision but more variable for Context Recall, indicating that relevant evidence was often surfaced without consistently covering all

reference-supporting information.

HyDE's high RAGAS Context Precision did not translate into consistent improvements in final answers. Within the evaluated questions, retrieval precision alone is therefore insufficient to favor HyDE over Hybrid; answer-quality metrics must also be considered.

V. Conclusions

For the models and conditions evaluated, MedGemma did not consistently outperform Gemma after grounded QLoRA adaptation. The study therefore does not establish a general advantage for the medically pretrained model once both models receive the same domain-specific supervision. QLoRA also showed no consistent benefit in Answer Correctness, despite improving Answer Relevancy under No RAG.

The main implication is that medical specialization and domain adaptation cannot substitute for evaluating the complete question-answering configuration. Hybrid retrieval produced more consistent correctness gains than QLoRA in this benchmark, but neither adaptation nor retrieval improved every assessed outcome. Configuration choices therefore require evidence from the intended task rather than assumptions based on medical specialization or fine-tuning alone.

These conclusions remain exploratory: they concern two model families, ten questions per scenario, and automated metrics, with relevant source documents available for retrieval. The comparison does not isolate the causal effect of medical pretraining, demonstrate model equivalence, or establish clinical reliability. Broader paired evaluation and clinician assessment are needed to determine whether these findings generalize.

VI. Limitations and Future Work

MaternaQA-es derives from Spanish-language obstetric documents and may not represent the full range of clinical questions or practice settings. With only 10 evaluated questions per scenario, the benchmark provides limited coverage of obstetric topics and source diversity. No clinician assessment has yet been conducted; automated metrics alone cannot establish clinical safety or usefulness. Future work should broaden question and source coverage and incorporate obstetric specialist assessment of answer correctness, safety, and clinical relevance before drawing conclusions about suitability for clinical use.

The approximate marginal intervals do not establish paired statistical significance and do not account for document clustering, subset-selection bias, or evaluator variability.

Future work will align reporting with TRIPOD-LLM [19] and Data Statements for NLP [20].

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